

Abhishek

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EDUCATION

Indian Institute of Technology Madras

Bachelor of Science in Data Science and Applications

Chennai, India

Sept. 2022 – Sept. 2026

EXPERIENCE

SignSetu | [Play Store](#)

Full Stack Developer

Chennai, India

Mar 2026 – Present

- Launched the **Connect app (Expo)** in Jun 2026: **4,100+** users and **80K+** game plays in four months.
- Own the **Turborepo** monorepo for the Next.js web app, API and mobile app, with **135 typed APIs**.
- Merged web and mobile accounts into one, migrating **140K records** across MongoDB and Postgres.
- Shipped v1.2.0 on **Expo SDK 57**, cutting p90 render time **27%**; added OTA and push for **1,000+ devices**.

Full Stack Developer Intern

May 2025 – Feb 2026

- Built **9 sign language games** played **92K+** times across web and mobile.
- Developed the **admin CMS** used to publish **559 lessons**, 1.8K vocab words and 8.5K media files.
- Wrote **29 Playwright** E2E tests covering login, lessons, games and admin CRUD, run in GitHub Actions.

GyanAlign | [gyanalign.com](#)

Software Development Engineer (Part-time)

Remote

Jun 2025 – Feb 2026

- Created a **legal research AI** over **42K+** **Supreme Court judgments** (1950–2026) and Indian statutes.
- Split documents by Section/Article into **1.4M+** **chunks**, stored in **pgvector** and searched via **Pinecone**.
- Designed an **18-node LangGraph agent** with reranking, citation checks and **Tavily** web fallback.
- Implemented **Razorpay subscriptions** with AI usage limits, student–recruiter matching and 7 user roles.

Amorcer (Acquired by 3Y Health) | [amorcer.com](#)

Lead Software Engineer (Part-time)

Remote, US

Jun 2025 – Dec 2025

- Built core features of the **virtual clinic platform** that 3Y Health acquired for its technology.
- Automated clinic website launch**: generate a Next.js site, push it to GitHub and deploy on **Vercel**.
- Used Pinecone search over **70+** **UI components** to pick the right sections for each site.

PROJECTS

Research Assistant Agent

[GitHub](#)

Next.js, FastAPI, LangGraph, PostgreSQL/pgvector, MCP, Yjs, gRPC, Docker

- Built a **LangGraph agent** that plans, drafts and critiques research roadmaps in up to 4 rounds.
- Turned papers into a **knowledge graph** of entities and relations (**GraphRAG**) on **pgvector**.
- Added a **Postgres job queue**, sandboxed code runs (E2B), **MCP** tools and live co-editing (Yjs).

Regional Dialect Synthesis Pipeline

[GitHub](#)

Python, PyTorch, Transformers, QLoRA/PEFT, Gemma 4 (GGUF), XTTS, FastAPI, Modal

- Fine-tuned **LLaMA 3.1 8B** and **Gemma 4** with **QLoRA** on 5,594 Hindi–Haryanvi pairs for translation.
- Fine-tuned **XTTS v2** for Haryanvi speech and served the full pipeline with FastAPI on Modal.

ACHIEVEMENTS

- Core developer of SignSetu, featured in **Forbes Accessibility 200 (2026)** and Purple Fest 2025.
- Finalist** in the **Google Code to Learn Competition**, among 25 students selected nationwide.
- Ranked in the **top 5.8%** on LeetCode with a **Knight** badge and peak rating of **1854**.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Go, C++, Java, SQL

Frameworks & AI: React, Next.js, React Native, Node.js, FastAPI, PyTorch, LangChain, LangGraph, QLoRA

Data & Cloud: PostgreSQL, MongoDB, Redis, Pinecone, AWS, GCP, Docker, Kafka, GitHub Actions